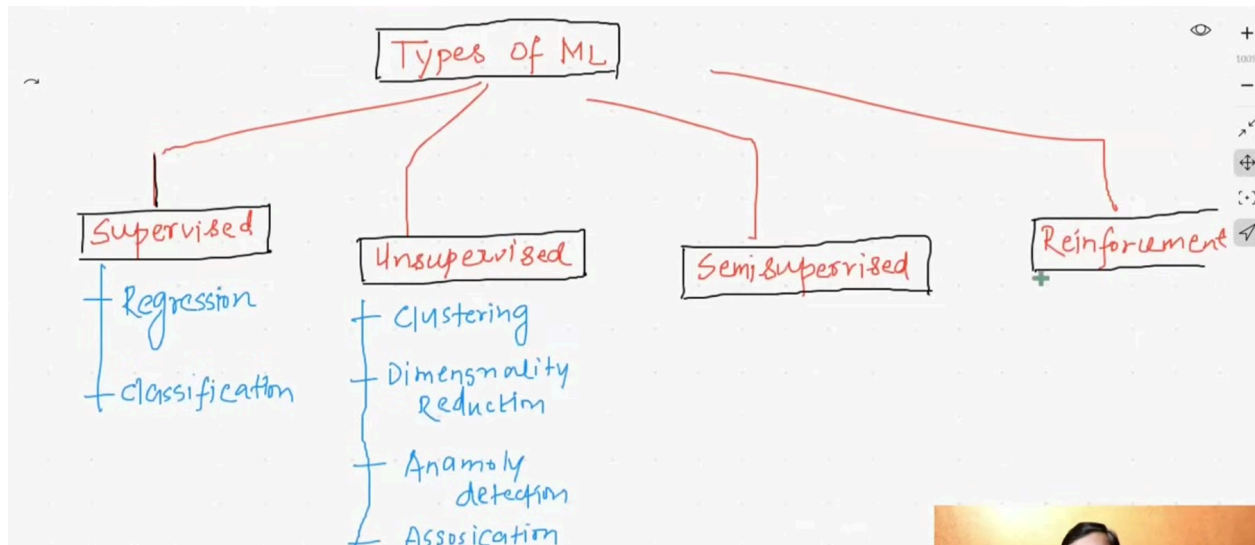


Types of ML



Supervised Learning

Definition

Supervised learning is a type of machine learning where the model learns from labeled data, meaning both input and correct output are already given.

Example

Predicting whether an email is spam or not spam using previously labeled emails.

Regression

Definition

Regression is a supervised learning technique used to predict continuous numerical values.

Example

Predicting the price of a house based on its size and location.

Classification

Definition

Classification is a supervised learning technique used to predict categories or class labels.

Example

Detecting whether a message is spam or not spam.

Unsupervised Learning

Definition

Unsupervised learning is a type of machine learning where the model learns from unlabeled data and finds hidden patterns or relationships in the data.

Example

Grouping customers based on their shopping behavior without predefined categories.

Clustering

Definition

Clustering is an unsupervised learning technique used to group similar data points into clusters.

Example

Grouping students based on their marks and study patterns.

Dimensionality Reduction

Definition

Dimensionality reduction is a technique used to reduce the number of input features while keeping important information.

Example

Reducing a dataset with 100 features into 10 important features for easier analysis.

Association Rule Learning

Definition

Association rule learning is a technique used to discover relationships or patterns between items in a dataset.

Example

In supermarkets, finding that customers who buy bread and butter also often buy milk.

Anomaly Detection

Definition

Anomaly detection is a machine learning technique used to identify unusual or abnormal data points that are different from normal patterns.

Example

Detecting fraudulent credit card transactions that do not match a user's normal spending behavior.

Semi-Supervised Learning

Definition

Semi-supervised learning is a type of machine learning that uses both labeled and unlabeled data for training.

Example

Using a small number of labeled images and many unlabeled images to train an image classification model.

Reinforcement Learning

Definition

Reinforcement learning is a type of machine learning where an agent learns by interacting with the environment and receives rewards or penalties based on its actions.

Example

Training a robot to walk by rewarding correct movements and penalizing wrong movements.

Batch Learning/ offline learning

Definition

Batch learning is a machine learning approach where the model is trained using the entire dataset at once. After training, the model is deployed, and it does not learn continuously from new data unless it is trained again.

Example

A company trains a sales prediction model using all sales data collected from the previous year. If new sales data arrives, the model must be retrained with the updated dataset.

Disadvantages of Batch Learning

1. **High Training Time**
Training on the complete dataset can take a long time, especially for large datasets.
2. **Requires Large Memory**
Since the whole dataset is used together, it needs high storage and memory.
3. **Cannot Learn Continuously**
The model cannot automatically learn from new incoming data.
4. **Retraining is Expensive**
Every time new data arrives, the entire model may need to be retrained.
5. **Not Suitable for Real-Time Systems**
It is not ideal for applications where data changes frequently, such as stock market prediction or online recommendations.

Online Learning

Definition

Online learning is a machine learning approach where the model learns continuously by updating itself with new data one step at a time.

Example

A recommendation system like YouTube continuously updates suggestions based on the user's latest watch history.

Disadvantages of Online Learning

1. **Sensitive to Noisy Data**
Incorrect or poor-quality data can reduce model accuracy.
2. **Risk of Wrong Updates**
Continuous updates may cause the model to learn incorrect patterns.
3. **Difficult to Maintain**
Monitoring and controlling the learning process can be complex.
4. **Requires Continuous Resources**
The system needs regular processing power and monitoring.
5. **Can Forget Old Patterns**
Sometimes the model may focus too much on recent data and ignore older useful information.

7. Batch Vs Online Learning

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Offline Learning	Features	Online Learning
Less complex as model is constant	 Complexity	Dynamic complexity as the model keeps evolving over time
Fewer computations, single time batch-based training	Computational Power	Continuous data ingestions result in consequent model refinement computations
Easier to implement	Use in Production	Difficult to implement and manage
Image Classification or anything related to Machine Learning - where data patterns remains constant without sudden concept drifts	Applications	Used in finance, economics, health where new data patterns are constantly emerging
Industry proven tools. E.g. Sci-kit, TensorFlow, Pytorch, Keras, Spark Mlib	Tools	Active research/New project tools: E.g. MOA, SAMOA, scikit-multiflow, streamDM

Instance-Based Learning

Definition

Instance-based learning is a machine learning approach where the model stores training examples and makes predictions by comparing new data with stored examples.

Example

K-Nearest Neighbor (KNN) predicts the category of a new data point by checking the closest stored data points.

Model-Based Learning

Definition

Model-based learning is a machine learning approach where the model learns patterns from the training data and creates a mathematical model for prediction.

Example

Linear Regression learns the relationship between house size and price to predict the price of a new house.